Syndromic Cholestasis in the Infant: Trisomy 21 and Trisomy 18, Arthrogryposis-Renal Dysfunction-Cholestasis Syndrome, Kabuki Syndrome and Smith-Lemli-Opitz Syndrome.

Picture a five-week-old boy with Down syndrome who is still jaundiced. His conjugated bilirubin, the fraction the liver has already processed but cannot excrete, is four milligrams per deciliter. He had a heart murmur at birth and an echocardiogram showed an atrioventricular septal defect. Everyone on the ward is comfortable with the explanation that babies with Down syndrome are often jaundiced. Now picture a six-week-old girl whose knees and wrists have been stiff since birth, whose skin is dry and scaly, who passes large volumes of dilute urine, and whose conjugated bilirubin is nine milligrams per deciliter while her gamma-glutamyl transferase, the enzyme usually called GGT, is normal. A surgical resident has booked a percutaneous liver biopsy for the next morning. And picture a three-day-old with low blood sugar, long eye openings and prominent finger pads, who is now becoming cholestatic.

The clinical question of this chapter is how to recognize the infant whose cholestasis is part of a syndrome, and what that recognition should change. The main idea is that the face, the hands, the feet, the skin, the joints, the kidneys, the blood sugar and the blood film are all part of the liver examination. When an infant looks different, the work-up stops being only a cholestasis work-up and becomes a syndrome work-up. That shift sends you to a karyotype, a sterol profile or a whole genome rather than to another liver panel. Just as important, two of the five conditions in this chapter change what you must not do, and three of them change what you must give.

A note on sources. The framework comes from the 2017 joint guideline on cholestatic jaundice in infants, written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. The European view of genetic cholestasis comes from the 2024 clinical practice guideline of the European Association for the Study of the Liver, known as EASL, led by Verkade. For Down syndrome we use the Stockholm cohort of Arnell and Fischler, published in 2012, the bilirubin study by Bahr and colleagues in 2020, a cholestasis series from Cairo by Kotb and colleagues in 2019, the prospective blood study by Roberts and colleagues in 2013, and the German prevention trial reported by Flasche and colleagues in 2018. For trisomy 18 we use the Ontario survival study by Nelson and colleagues in the Journal of the American Medical Association in 2016 and the heart surgery study by Kosiv and colleagues in 2017. For the arthrogryposis syndrome we use the gene discoveries by Gissen and colleagues in 2004 and Cullinane and

colleagues in 2010, and the large clinical series by Gissen and colleagues in 2006. For Kabuki syndrome we use the 2024 hyperinsulinism series by Rosenfeld and colleagues, and for Smith-Lemli-Opitz syndrome the GeneReviews chapter by Nowaczyk and Wassif and the liver series by Rossi and colleagues in 2005.

Part one. Five syndromes, five different first tests.

Start with a map. Trisomy 21, or Down syndrome, is the commonest syndrome you will see with neonatal cholestasis. The clue is obvious, and the first test that matters is not a liver test at all but a blood film, looking for the immature white cells called blasts. Trisomy 18, also called Edwards syndrome, shows clenched fists with overlapping fingers, rocker-bottom feet and growth restriction before birth. The first test is a rapid test for an extra chromosome, followed by a karyotype, and what it changes is the conversation with the family about goals of care. The third condition has a long name, arthrogryposis, renal dysfunction and cholestasis syndrome, which everyone shortens to ARC. Arthrogryposis means joints that have been fixed in one position since before birth. The genes are VPS33B and VIPAS39. The clue is contractures, dry scaly skin called ichthyosis, and a lot of urine. The GGT is low or normal. The first tests are urine studies and sequencing, and what changes is that you must not do a percutaneous biopsy. Kabuki syndrome, caused by changes in the genes KMT2D or KDM6A, shows long eye openings, arched eyebrows and persistent fetal pads on the fingertips; the first test is a glucose with an insulin level at the moment the sugar is low, and what changes is that some of these babies need the drug diazoxide. Smith-Lemli-Opitz syndrome, which we will call SLOS, is caused by the gene DHCR7. Look for webbing between the second and third toes and for genital anomalies. The GGT is again low or normal, the first test is plasma 7-dehydrocholesterol, and the treatment is cholesterol.

One rule ties several of these together. When conjugated bilirubin is high but GGT is low or normal, the list narrows to a short set: ARC syndrome, SLOS, the inborn errors of bile acid synthesis, and the progressive familial intrahepatic cholestasis types. These are the conditions in which the canalicular membrane, the pump machinery or the bile acids themselves are abnormal, so the bile duct cells that make GGT are never injured.

Part two. Trisomy 21: jaundice is common, cholestasis is not.

The word jaundice hides two different problems in Down syndrome. The first is ordinary unconjugated jaundice, which is very common. In the study by Bahr and colleagues, 357 newborns with Down syndrome were compared with more than three hundred and seventy thousand controls. About twenty-three and a half percent of the babies with Down syndrome had a total bilirubin above the ninety-fifth centile, compared with five percent of controls. Sixty-two percent of them received phototherapy, compared with seven percent of controls.

That is a red cell and conjugation story, and it is not what this chapter is about.

Cholestasis is much less common. In Stockholm between 2005 and 2011, Arnell and Fischler followed 206 newborns with Down syndrome, about one in every eight hundred and eighty births. Eight of them, three point nine percent, developed cholestasis. Every one of those eight had congenital heart disease. And the three infants with severe cholestasis were exactly the three who also had disease of the bone marrow, the condition called transient abnormal myelopoiesis. Two of those three died and the third developed cirrhosis. So in a population setting, cholestasis in Down syndrome is uncommon, it travels with heart disease, and when it is severe it is usually a marrow disease that has reached the liver.

Referral series look different, because they collect the sicker and more persistent cases. In a Cairo series, 61 of 779 cholestatic infants, almost eight percent, had Down syndrome. Among the 55 who were followed, none had extrahepatic biliary atresia. About two thirds had a neonatal hepatitis picture and one third had a paucity of bile ducts on biopsy without the Alagille syndrome. Almost two thirds resolved. No population study has shown that Down syndrome raises the risk of biliary atresia, so the practical message is not that atresia is impossible, but that the atresia work-up proceeds exactly as it would in any other infant.

Part three. Transient abnormal myelopoiesis: a fetal leukemia that scars the liver.

Transient abnormal myelopoiesis, often shortened to TAM, is the reason a hepatologist must think like a hematologist in Down syndrome. The mechanism unfolds in five steps. First, the extra copy of chromosome 21 expands the progenitor cells that make platelets and red cells in the fetal liver, which is where blood is made before birth. Second, a somatic mutation, meaning one acquired in those cells rather than inherited, hits a gene called GATA1. GATA1 is a master switch for platelet and red cell development. The mutation produces a shortened protein, and blast cells start to multiply. Third, these blasts, many of them early platelet precursors called megakaryoblasts, fill the sinusoids of the liver. Growth factors released by these cells are thought to drive scarring, so the liver can become fibrotic within weeks. Fourth, the baby shows a large liver, ascites, a bleeding tendency and cholestasis, and in the worst cases hydrops, which is fluid throughout the body before birth. Fifth, the story has two endings. In most babies the abnormal clone disappears by about three months of age. In roughly one in five, a true myeloid leukemia of Down syndrome develops later, usually in the first four years.

How common is the GATA1 clone? Roberts and colleagues studied 200 unselected newborns with Down syndrome. Almost all of them, ninety-seven and a half percent, had some blasts in the blood. Seventeen, eight and a half percent, had a GATA1 mutation detectable by standard

sequencing, all with blasts above ten percent. When the researchers used a much more sensitive method, next-generation sequencing, they found small clones in about one in five of the remaining babies tested. These are called silent clones because they cause no symptoms. A later thesis from the same group found that a persistently enlarged liver was associated with progression to leukemia.

The treatment evidence comes from the German prevention trial, called TMD Prevention 2007, reported by Flasche and colleagues. One hundred and two children with symptomatic TAM received low-dose cytarabine, a chemotherapy drug used at a fraction of its leukemia dose. Compared with historical controls, early death fell from thirty-three percent to twelve percent. Liver fibrosis was seen in seven of 104 historical controls and in none of the 102 trial patients. What cytarabine did not change was the later leukemia, which still occurred in about a quarter of children at five years, compared with twenty-two percent historically. In other words, treatment rescues the liver and the newborn, but not the clone. The risk factors for early death were a high white count, ascites, liver dysfunction, hydrops and prematurity, which is almost a list of what a hepatologist sees.

So the practical rule is simple. Ask for a blood film in every cholestatic infant with Down syndrome. Blasts above ten percent with a large liver and ascites mean TAM until proven otherwise, and hematology should be called the same day.

Part four. The rest of the list in Down syndrome.

Not every jaundiced baby with Down syndrome has TAM, and the other contributors are worth knowing as numbers. Congenital hypothyroidism occurs in about one percent of babies with Down syndrome, roughly twenty-eight times the general rate, with about seven in a thousand permanent and three in a thousand transient. In the large American Down syndrome projects, involving almost nineteen hundred infants, duodenal atresia or stenosis occurred in about four percent and Hirschsprung disease in less than one percent. Each brings surgery, fasting and parenteral nutrition, and Hirschsprung brings the risk of enterocolitis and sepsis. Congenital heart disease was present in almost half of the Stockholm cohort, and low cardiac output and venous congestion add their own injury to the liver. Most cholestasis in Down syndrome is therefore multifactorial and self-limited. The trap is to attribute it to the syndrome without sending a blood film, without checking the thyroid, and without following the stool color.

Part five. Trisomy 18: the liver is rarely the decision.

Trisomy 18 occurs in about one in six thousand births. The survival data frame everything else. In the Ontario study of 254 children, the median survival was nine days. About twelve and a half percent were alive at one year and about ten percent at ten years. Fourteen percent had surgery, and among those who did, two thirds were alive one year after the first operation,

which reflects both the benefit of surgery and the fact that surgeons select the strongest babies. A United States study of more than a thousand infants with trisomy 18, nine in ten of whom had heart disease, found in-hospital death in sixteen percent of those who had heart surgery and forty-four percent of those who did not, again with large selection bias.

Liver disease in trisomy 18 is described mostly in case reports, including extrahepatic biliary atresia. The best-documented liver association is cancer. In a review of 45 malignancies reported in trisomy 18, 29 were liver cancers, mostly hepatoblastoma, and unusually they occurred mostly in girls, the reverse of the usual male predominance of hepatoblastoma. In practice, the decision for a cholestatic infant with trisomy 18 is rarely driven by the liver. It is a family conversation about goals of care, informed by honest survival figures, in which liver investigations are chosen for what they would change.

Part six. ARC syndrome: a trafficking defect in four organs.

ARC syndrome is the condition in this chapter that most rewards understanding the mechanism. Every epithelial cell has two sides. The apical side faces a lumen, such as the bile canaliculus or the kidney tubule, and the basolateral side faces the blood. Proteins have to be delivered to the correct side, and the delivery depends on a sorting station called the apical recycling endosome. The protein VPS33B belongs to a family of regulators of membrane fusion. It binds a partner, VIPAS39, which used to be called VIPAR, and a small transport protein called RAB11A. When either partner is missing, apical membrane proteins end up in the wrong place, and the tight junctions that seal cells together become abnormal.

Now follow that one defect through four organs. In the liver, the canalicular transport machinery is not properly delivered, and the result is severe cholestasis with a normal or low GGT. In the kidney, the proximal tubule cannot reabsorb what it should, which produces renal Fanconi syndrome, with loss of glucose, phosphate, bicarbonate and amino acids in the urine, and renal tubular acidosis. In the skin, failed epithelial polarity produces ichthyosis. And in platelets, VPS33B is needed to form the alpha granules, the storage packets that platelets release when they activate. Without them, platelets look pale and large, like those of grey platelet syndrome, and they do not work properly even when the count is normal.

The genetics are straightforward. In the largest series, mutations in VPS33B were found in 28 of 35 families, about three quarters, and VIPAS39 accounts for most of the rest. The clinical picture follows the mechanism. In the 2006 series of 62 children, all had severe failure to thrive and 60 had arthrogryposis. About one in ten had a structural heart defect. In an earlier British series of six children, every child had severe cholestasis with a normal GGT, five had renal Fanconi syndrome, and the sixth had nephrogenic diabetes insipidus instead, meaning the kidney could not concentrate urine. Large platelets and ichthyosis were common. Most

children in the classic series died in the first year of life, many within six months.

Part seven. The biopsy that can kill.

This is the single most important practical point in the chapter. In the 2006 series, sixteen children had a diagnostic biopsy of the liver or the kidney, and seven of them had a life-threatening hemorrhage. Of the eleven children who had severe bleeding, nine had a normal platelet count and normal platelet appearance on the blood film. The count reassures no one, because the defect is in platelet function, not number. The rule is therefore absolute: if you suspect ARC, do not perform a percutaneous biopsy. Make the diagnosis with sequencing, with urine studies showing tubular loss, and with platelet studies such as electron microscopy or flow cytometry for the alpha granule defect.

There is a milder end to the spectrum. A 2022 review collected nineteen long-term survivors, aged between three and fifty-nine years at last report, and eighteen of them carried VPS33B variants. Missense variants, which change a single amino acid rather than destroying the protein, are associated with milder disease, and one such variant in Chinese children caused isolated low-GGT cholestasis without the other features. These attenuated patients are the ones who eventually need procedures. When surgery or a biopsy is unavoidable, plan it with hematology; platelet transfusion and antifibrinolytic drugs have been used.

Part eight. When the GGT is low, look beyond the liver.

Because ARC and SLOS both present with low or normal GGT, it is worth laying them beside the other low-GGT causes and asking which extrahepatic sign separates them. ARC has contractures, Fanconi syndrome, ichthyosis and large platelets, and the action is to avoid biopsy. SLOS has toe webbing, genital anomalies and a small head, and the action is cholesterol. The inborn errors of bile acid synthesis are not syndromic; they show fat-soluble vitamin deficiency, and the test is urine bile acids by mass spectrometry, because cholic acid treatment works. Progressive familial intrahepatic cholestasis type one, caused by ATP8B1, brings diarrhea, pancreatitis and deafness, and the diarrhea can worsen after transplant. Type two, caused by ABCB11, has no extrahepatic sign but a high alanine aminotransferase and an early risk of liver cancer. Tight junction protein 2 deficiency is another low-GGT cause with a cancer risk. A useful pattern is this: low GGT with low or normal serum bile acids points to a synthesis defect, while low GGT with high serum bile acids points to a transport or trafficking defect.

Part nine. Kabuki syndrome: the jaundiced, hypoglycemic, dysmorphic baby.

Kabuki syndrome affects about one in thirty-two thousand people. Between fifty-five and eighty percent have a variant in KMT2D, and two to six percent have a variant in KDM6A, a gene on the X chromosome. Both genes encode enzymes that modify chromatin, the packaging of DNA, so the syndrome is a disorder of gene regulation in many tissues.

The liver involvement is sporadic rather than characteristic. Reported problems include biliary atresia, among them a cystic form in a child with a new KDM6A variant, neonatal sclerosing cholangitis and hepatic fibrosis. No cohort gives a reliable frequency. The practical point is that a pattern of high GGT with pale stools still demands the atresia pathway, whatever the face suggests.

The more important link for the neonatal hepatologist is hyperinsulinism. In the 2024 series of 33 children with Kabuki syndrome and hyperinsulinism, about three quarters had KMT2D and fifteen percent had KDM6A, which is more than expected from its overall share. Three quarters were hypoglycemic on the first day of life. Ninety-two percent responded to diazoxide, a drug that keeps the insulin-secreting cell from firing. And in sixty-one percent, hyperinsulinism was diagnosed before Kabuki syndrome itself: the low sugar came first and the face came second. Among 449 children with confirmed Kabuki syndrome, the estimated prevalence of hyperinsulinism was about four percent. The trap is to assume that every hypoglycemic, cholestatic infant has hypopituitarism. Hypopituitarism is the classic cause and must be excluded with cortisol and thyroid tests, but you should also draw insulin, beta-hydroxybutyrate and free fatty acids at the moment the glucose is low. A detectable insulin with suppressed ketones points to hyperinsulinism.

Part ten. Smith-Lemli-Opitz syndrome: the last step to cholesterol is missing.

The mechanism of SLOS is a single missing enzyme at the end of the cholesterol pathway. The enzyme 7-dehydrocholesterol reductase, made by the DHCR7 gene, converts 7-dehydrocholesterol, usually shortened to 7-DHC, into cholesterol. When it fails, 7-DHC and a related sterol called 8-DHC accumulate, and cholesterol falls, although not always below the normal range. Cholesterol is needed for a developmental signal called hedgehog signaling, which shapes the limbs, the genitals, the face and the brain; that is why the syndrome includes toe webbing, genital anomalies, cleft palate and a small head. Cholesterol is also the raw material for bile acids, and the abnormal sterols may be toxic to the liver, so cholestasis can occur, sometimes with a normal GGT. The treatment follows directly: dietary cholesterol, with bile acids in cholestatic infants, and the neonatal cholestasis often resolves.

How often is the liver involved? Reported figures range from two and a half to six percent of patients with SLOS, but one Italian center found liver disease in sixteen percent of its cohort.

Rossi and colleagues described two patterns: progressive cholestasis with a normal GGT, which was linked to early death, and isolated stable elevation of the transaminases, which was milder. Simvastatin was tried as a way to lower 7-DHC in a trial, but GeneReviews does not recommend it; cholesterol remains the treatment.

The diagnostic trap is the cholesterol level itself. SLOS affects about one in forty thousand births in North America, and about one percent of people carry a variant. But total cholesterol is normal in about ten percent of patients, so a normal cholesterol does not exclude the diagnosis. The test is plasma sterol analysis by gas chromatography with mass spectrometry, looking for high 7-DHC, with fibroblast sterols if the result is borderline. About twenty-eight percent of alleles carry one particular splice variant that destroys the enzyme and gives severe disease. Send the sterols before cholesterol-rich feeds or a transfusion confuse the picture. Mortality is highest when cholesterol is around ten milligrams per deciliter.

Part eleven. The mimics and the genetics.

Several other syndromes reach the liver clinic. Alagille syndrome, caused by JAG1 and less often NOTCH2, is the commonest familial intrahepatic cholestasis and has its own block in this course. Biliary atresia splenic malformation, often called BASM, made up about ten percent of 548 atresias in the British series by Davenport and colleagues. It comes with polysplenia, situs inversus and heart defects, and maternal diabetes was present in twelve and a half percent of these mothers compared with just over one percent in isolated atresia. Native-liver survival was forty-six percent at five years. In the North American ChiLDReN cohort of 289 atresias, eighty-four percent were isolated, six percent had a major malformation without a laterality defect, and ten percent had laterality defects. Trichohepatoenteric syndrome, caused by TTC37 or SKIV2L, combines intractable diarrhea, woolly hair and immune deficiency, with liver disease in about half. NISCH syndrome, caused by CLDN1, combines ichthyosis, scalp hair loss and sclerosing cholangitis, with jaundice in eighty-four percent of the nineteen reported patients. Zellweger spectrum disorders, most often caused by PEX1, bring hypotonia, seizures and kidney cysts, and the first test is very-long-chain fatty acids.

Now to genetic testing, because the choice of test matters. A cholestasis gene panel is designed around liver genes. In the largest study, a sixty-six-gene panel in more than two thousand cholestatic patients found a diagnosis in twelve percent. An eighteen-gene panel in 109 Japanese infants found a diagnosis in twenty-six percent overall, rising to seventy-one percent when a genetic cause was suspected clinically, which shows that pre-test probability drives yield. Exome sequencing in a Chinese cholestatic subgroup found a cause in forty-six percent. Rapid whole-genome sequencing in critically ill infants diagnosed fifty-seven percent of 35 infants in the first study, called STAT-seq, and sixty-five percent of those diagnoses changed

management. In the larger NSIGHT2 study, ultra-rapid genome sequencing diagnosed forty-six percent with results in about two days. The American College of Medical Genetics and Genomics, in 2021, recommended exome or genome sequencing as a first- or second-tier test for children with congenital anomalies. And a trisomy is found with a rapid test for aneuploidy, a karyotype or a chromosomal microarray, not with a sequencing panel.

Part twelve. The approach in the first seventy-two hours.

Here is a sequence that runs alongside the standard cholestasis algorithm, not instead of it. On day one, at the bedside, examine the face, hands, feet, skin, genitals and joints, look at the stool color, listen for a murmur and feel for the spleen. Also on day one, send blood and urine: GGT, glucose with insulin, thyroid-stimulating hormone, cortisol, a blood film for blasts, plasma 7-DHC, very-long-chain fatty acids, and urine for glucose, phosphate and amino acids to look for Fanconi syndrome. On the same day, apply two safety rules: if there are contractures or tubular loss, no percutaneous biopsy; if there are blasts, call hematology now. On day two, image: an ultrasound looking for signs of atresia and for the spleen, an echocardiogram, and a renal ultrasound. Between day two and day seven, send a rapid aneuploidy test, then trio genome or exome sequencing with analysis of copy-number changes, meaning deleted or duplicated segments of chromosomes.

Two principles protect the child. First, do not let the syndrome delay the atresia work-up; pale stools still mean a cholangiogram early enough to allow a Kasai operation before about sixty days of age, whatever the face looks like. Second, send all the safe blood and urine tests together on day one. The genome result usually arrives after decisions about biopsy and surgery, so those decisions must be planned with the result still pending.

Part thirteen. What the guidelines say.

The 2017 joint North American and European guideline lists several syndromic causes in its first table, including ARC, Alagille syndrome, trisomy 21 and NISCH, and advises a low threshold for gene panels or exome sequencing. The 2024 EASL guideline on genetic cholestatic diseases focuses on alpha-1 antitrypsin deficiency, Alagille syndrome, the progressive familial intrahepatic cholestasis types and bile acid synthesis defects. It mentions VPS33B only in a list of rarer genes. So the two documents differ on how prominently the syndromes appear. They are aligned on timing: EASL recommends early genetic testing with a panel or exome, moving to genome sequencing if the result is negative. EASL adds that negative or uncertain results should be re-analyzed at least every three years, a point the 2017 guideline does not address, so they differ there too.

There are three gaps. Neither liver guideline mentions a karyotype or microarray, although the genetics society recommends exome or genome as a first-tier test in congenital anomalies.

Neither mentions Smith-Lemli-Opitz syndrome. And neither addresses transient abnormal myelopoiesis, which lives in the hematology literature. There is no AASLD guidance, meaning the American Association for the Study of Liver Diseases, on syndromic infant cholestasis. The material gap is the biopsy. Neither liver guideline warns against biopsy in suspected ARC, yet seven of sixteen biopsied children in the largest series had life-threatening bleeding. That warning lives only in case series, which means you have to carry it in your head.

Part fourteen. Pearls and traps.

The common errors fall into four groups. The first is saying that it is just Down syndrome: no blood film is sent, TAM fibrosis is missed, and the atresia work-up is skipped. The second is booking a biopsy for low-GGT cholestasis without examining the joints, being reassured by a normal platelet count, and then facing a hemorrhage. The third is using total cholesterol as the screen for SLOS, never sending 7-DHC, or sending it after a transfusion. The fourth is sending a liver panel when the child needs a genome, so that syndromic genes and copy-number changes are never examined and the result is never re-analyzed. And one more: labeling a hypoglycemic, cholestatic infant as hypopituitary when insulin was never measured.

The numbers worth carrying are these. Cholestasis affects about four percent of newborns with Down syndrome, and the severe cases are TAM. GATA1 clones are found in about eight and a half percent by standard sequencing and in about a fifth more by sensitive sequencing. Low-dose cytarabine cut early death from thirty-three to twelve percent and abolished liver fibrosis in the trial. In trisomy 18, median survival is nine days and about one in eight children is alive at one year. In ARC, almost every child has arthrogryposis, and seven of sixteen biopsied children bled dangerously. In Kabuki syndrome, about four percent have hyperinsulinism and nine in ten of them respond to diazoxide. In SLOS, cholesterol is normal in one patient in ten, so send 7-DHC. BASM is a tenth of biliary atresia. And rapid genome sequencing in sick infants finds a diagnosis in roughly forty to fifty-seven percent, compared with twelve to twenty-six percent for a liver panel alone.

What to do differently in clinic.

Let us return to the three babies at the start. The five-week-old with Down syndrome and heart disease needs a blood film, a thyroid test and the standard atresia pathway, including attention to stool color; if he had had blasts, ascites and a large liver in the first days of life, the call to hematology and low-dose cytarabine would have been the urgent step. The six-week-old with contractures, ichthyosis, polyuria and a normal GGT has ARC syndrome until proven otherwise, and the biopsy should be cancelled. Send urine studies, platelet studies and a rapid genome instead. The three-day-old with low sugar and Kabuki features needs a critical sample with insulin, a trial of diazoxide if hyperinsulinism is confirmed, and sequencing that covers

KMT2D and KDM6A.

So, in clinic tomorrow: examine the whole baby before ordering the next liver test. In every cholestatic infant with Down syndrome, send a blood film. When the GGT is low, look at the joints, the skin and the urine before anyone books a biopsy. When cholesterol is normal but the toes are webbed, send 7-DHC. When glucose is low, measure insulin as well as cortisol. And when a dysmorphic infant has a negative cholestasis panel, move to trio genome sequencing with copy-number analysis, and plan to re-analyze it within a few years.